Bioinformatics Bootcamp 2026

Assessment # 02

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PART-1

Core Challenge

Modern biological research depends on efficiently retrieving, interpreting, and integrating information from multiple biological databases. Selecting the appropriate database is often the first and most critical step in solving a biological research problem.

<u>Question 1:</u>

A molecular biologist has been provided with only a gene accession number from a published research article and needs to retrieve the corresponding DNA sequence, protein sequence, associated literature, and gene information. Describe the step-by-step approach you would follow using NCBI resources. Explain which databases and tools you would access during this process and why.

<u>Answer:</u>

If I was provided with just a gene accession number, I would go to the NCBI website and enter the number into the search bar. The first thing you need to do is determine what type of record the accession number is.

I would go to the Gene database and get more detailed information about the gene, such as the official name, location on the chromosome, function, etc., and then organisms that the gene is related to. This database gives you a single source of information about the gene and connects you to many other databases.

I would then go to the Nucleotide database to get the whole DNA sequence. This database contains genomic and transcript sequences and it's possible to download the sequence in FASTA format for further analysis.

I would click on the protein links in the gene record and then open the Protein database to get the protein sequence. Protein sequences and functional annotations are included in this database and are translated.

When looking for scientific publications about the gene, I would use PubMed, which would give me access to research literature, reviews, and experimental studies. These publications enable the knowledge of findings and biological significance of the past.

Last but not least, I would employ more NCBI tools like BLAST to analyse the sequence against known sequences and find similarities. I can get comprehensive information about the gene, its sequence, function and supporting literature using Information from Gene, Nucleotide, Protein databases and PubMed.

Question 2:

A structural biologist is interested in studying the three-dimensional structure of the human insulin protein to understand how mutations may affect its function. Which database would you recommend for this purpose? Explain how you would retrieve the structure and describe the information available within the database that would support the research.

Answer:

The Protein Data Bank (PDB) is the recommended database for studying the three-dimensional structure of human insulin, as it is the most important database of experimentally determined protein structures.

I would go to the PDB website and type in "insulin" (or any known insulin PDB ID). The search results give you available structures, which have been determined by various techniques, including X-ray crystallography, NMR spectroscopy or cryo-electron microscopy.

Once I have chosen the protein I want to see, I am able to display the structure in an interactive 3D viewer. This enables rotation, zooming and viewing of key structural areas. In addition to protein chains, amino acid residues, ligands, structural resolution, and experimental methods used to determine the structure information, the database contains information about protein chains, amino acid residues, ligands, structural resolution, and experimental methods used to determine the structure.

The PDB is particularly helpful when dealing with mutations as it allows the researcher to know which amino acids are in important functional regions. Comparisons can be used to provide information on the structure of the protein and how an alteration in amino acid sequence could affect its folding, stability, and/or biological activity.

Hence PDB is the best choice of database as it contains structured information that is not available in sequence databases. The knowledge is crucial for the understanding of the structure-function relationship of proteins.

Question 3:

Imagine you have been assigned a protein with no prior knowledge except its accession ID. Design a complete workflow describing how you would retrieve, analyze, and interpret all available information about this protein using NCBI, UniProt, and the Protein Data Bank (PDB). Explain the purpose of each step and how the information from one database complements the others.

Answer:

If I'm provided with a protein accession ID only I would use the NCBI, UniProt, and PDB in a systematic way.

A search of the accession number in the NCBI Protein database would be my first step. This will give you the protein sequence, source organism, and links to other information relating to the gene. I can find out from the linked Gene database what the biological function of the gene, where it is in the genome, and what it is known to do.

Then, I would compare the protein sequence with other proteins using BLAST. This aids in identification of homologous proteins, conserved regions, and potential functions by comparison with known proteins.

After obtaining the basic information, I would search the protein in UniProt. The UniProt annotations include high-quality functional and structural data, such as protein function, domains, active sites, pathways, subcellular localization and disease associations. It also includes cross references to numerous external databases.

The next step would be to look for the protein in the Protein Data Bank (PDB). I can learn about the 3D organization of a structure and recognize key structural features if a structure is available. When no structure is available, UniProt is likely to be connected to predicted structural resources.

I will lastly merge the data from all three databases. NCBI offers sequence and gene level data, UniProt offers detailed functional annotations and PDB offers structural insights. These resources offer a comprehensive view of the protein sequence, function, evolution and structure, facilitating correct biological interpretation.

PART-2

Sequence Retrieval Exercise

Choose one biological sequence (DNA or Protein) from NCBI, UniProt, or the Protein Data Bank (PDB).

NCBI

insulin, isoform 2 precursor [Homo sapiens]

NCBI Reference Sequence: NP_001035835.1

Identical Proteins FASTA Graphics

Go to:

```
LOCUS       NP_001035835             200 aa            linear   PRI 27-APR-2025
DEFINITION  insulin, isoform 2 precursor [Homo sapiens].
ACCESSION   NP_001035835
VERSION     NP_001035835.1
DBSOURCE    REFSEQ: accession NM_001042376.3
KEYWORDS    RefSeq.
SOURCE      Homo sapiens (human)
  ORGANISM  Homo sapiens
            Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
            Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini;
            Catarrhini; Hominidae; Homo.
```

- **Accession ID**

NP_001035835.1

- **Protein Name**

Insulin, Isoform 2 Precursor

- **Source Organism**

Homo sapiens (Human)

- **Database Used**

NCBI Protein Database

Sequence:

>NP_001035835.1 insulin, isoform 2 precursor [Homo sapiens]

MALWMRLLPLLALLALWGPDPAAAFVNQHLCGSHLVEALYLVCGERGFFYTPKTRR
EAEDLQASALSLSSSTSTWPEGLDATARAPPALVVTANIGQAGGSSSRQFRQRALGTS
DSPVLFIHCPGAAGTAQGLEYRGRRVTTELVWEEVDSSPQPQGSESLPAQPPAQPAPQ
PEPQQAREPSPEVSCCGLWPRRPQRSQN

Brief Description

Insulin isoform 2 precursor is a protein encoded by the human INS gene. It is initially produced as a precursor molecule that undergoes several processing steps to generate mature insulin, a hormone essential for maintaining normal blood glucose levels. Insulin plays a critical role in regulating carbohydrate, fat, and protein metabolism by facilitating the uptake of glucose from the bloodstream into body cells.

The biological function of insulin is to help control blood sugar levels and ensure that cells receive the energy needed for normal physiological processes. When insulin production is reduced or when cells become resistant to its effects, disorders such as diabetes mellitus can develop. Therefore, insulin is one of the most extensively studied proteins in biomedical research.

This protein is important because it is directly involved in human health and disease. Understanding its structure, sequence, and function has contributed significantly to the development of insulin therapies used worldwide for diabetes treatment.

I selected the **NCBI Protein Database** because it provides reliable protein sequence information along with links to related genes, scientific literature, conserved domains, and other biological resources. The database is easy to access and integrates information from multiple sources, making it a valuable tool for studying proteins and their biological roles.